Wireless Networking

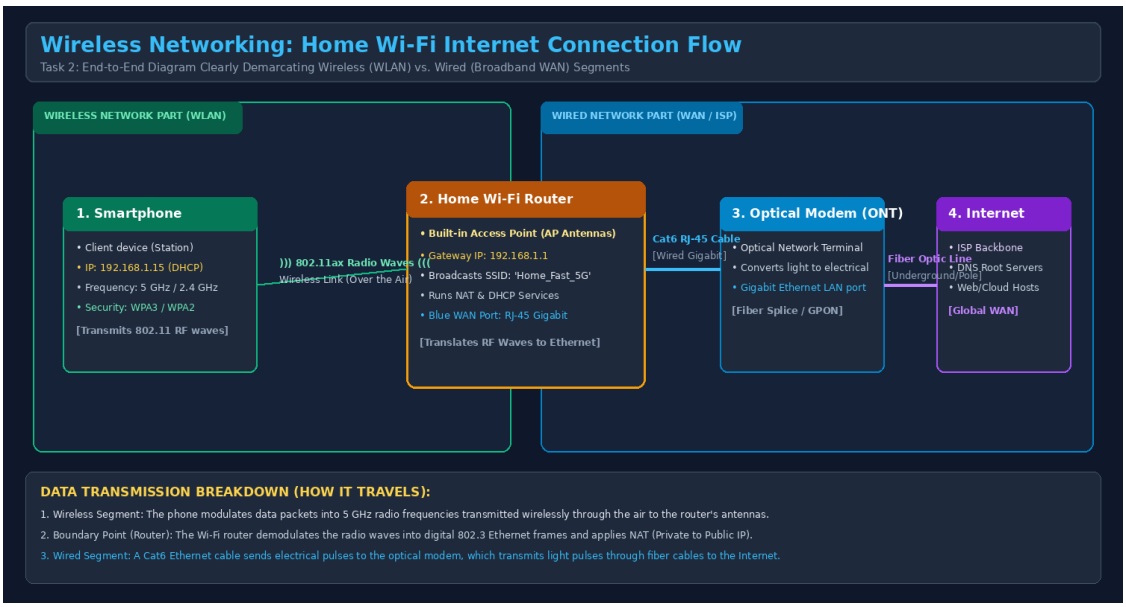
Task 1: 3 Popular Wi-Fi Standards Comparison

Wi-Fi Standard	Industry Marketing Name	Key Differences in Maximum Speed & Frequency Band
IEEE 802.11n	**Wi-Fi 4**	• **Speed:** Up to **600 Mbps** (using 4×4 MIMO).

• **Frequency Bands:** Dual-band support (**2.4 GHz2.4\text{ GHz}2.4 GHz and 5 GHz5\text{ GHz}5 GHz**). Introduced MIMO (Multiple Input, Multiple Output) multi-antenna transmission. | | **IEEE 802.11ac** | **Wi-Fi 5** | • **Speed:** Up to **6.9 Gbps6.9\text{ Gbps}6.9 Gbps** (with wider 80 MHz/160 MHz80\text{ MHz} / 160\text{ MHz}80 MHz/160 MHz channels and 256-QAM).

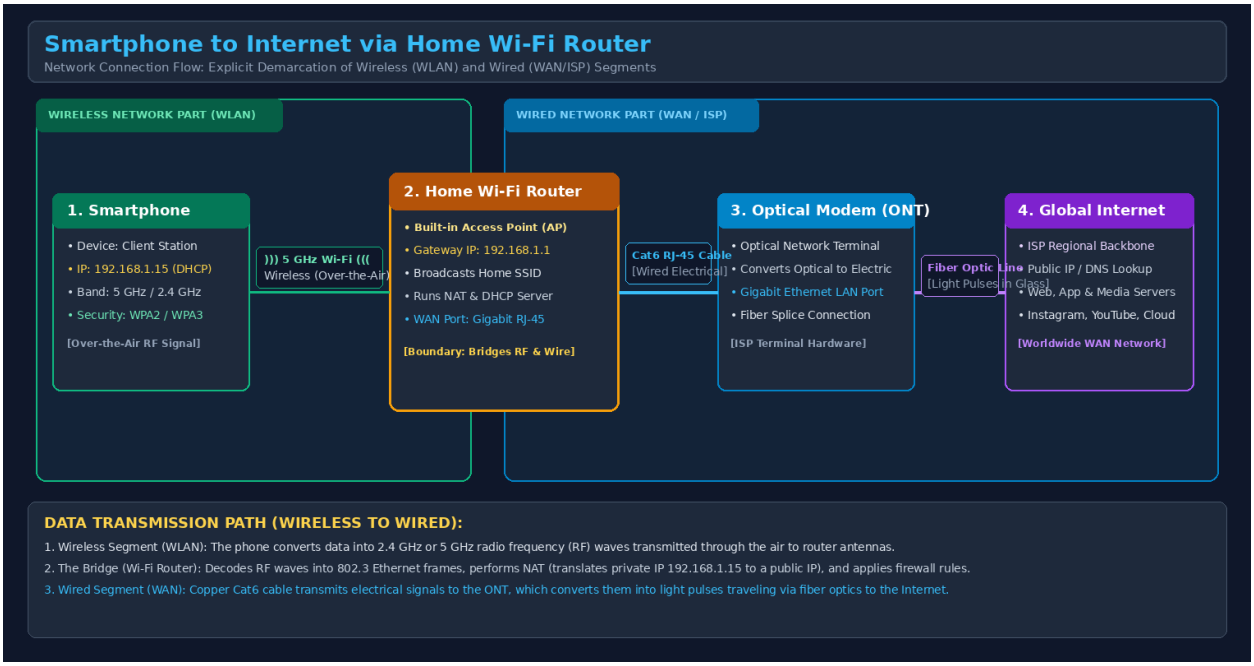
• **Frequency Bands:** Operates **strictly on the 5 GHz5\text{ GHz}5 GHz band** for data (uses legacy 2.4 GHz2.4\text{ GHz}2.4 GHz only for backward compatibility). | | **IEEE 802.11ax** | **Wi-Fi 6 / 6E** | • **Speed:** Up to **9.6 Gbps9.6\text{ Gbps}9.6 Gbps**.

• **Frequency Bands:** Operates across **2.4 GHz2.4\text{ GHz}2.4 GHz, 5 GHz5\text{ GHz}5 GHz, and 6 GHz6\text{ GHz}6 GHz** (Wi-Fi 6E). Introduces **OFDMA** (Orthogonal Frequency Division Multiple Access) and **MU-MIMO**, allowing routers to communicate with dozens of devices simultaneously without congestion. |



Task 2: Home Wi-Fi Network Connection Diagram

The diagram below maps how a smartphone connects to the Internet through a home router, clearly distinguishing the wireless over-the-air path from the physical wired infrastructure:



Connection Details:

1. **Wireless Network Segment (WLAN):**
    - The **Smartphone** communicates with the **Wi-Fi Router's antennas** over the air using high-frequency RF radio waves (5 GHz5\text{ GHz}5 GHz or 2.4 GHz2.4\text{ GHz}2.4 GHz, IEEE 802.11ax). No physical cables are attached to the phone.
2. **The Boundary / Bridge (Home Wi-Fi Router):**
    - The router receives the radio waves, converts them into digital Ethernet frames, and performs **NAT (Network Address Translation)** to swap the phone's private IP (192.168.1.15) with the broadband public IP.
3. **Wired Network Segment (Broadband WAN / ISP):**
    - **Cat6 Ethernet Patch Cable:** Connects the router's blue **WAN port** to the **Optical Network Terminal (ONT modem)** using copper RJ-45 connectors.
    - **Fiber Optic Cable:** Sends data as pulses of light through underground glass fibers to the Internet Service Provider's core data center and root DNS servers.

Task 3: Wi-Fi Network Scan & Selection Analysis

A live Wi-Fi scan was conducted using a mobile phone / laptop network analyzer. Below are three observed networks:

Network SSID	Frequency Band	Signal Strength (RSSI / Bars)	Security Protocol	Network Quality
Home_Fiber_5G	5 GHz	**−42 dBm (Excellent / Full 4 bars)**	**WPA3-Personal (SAE)**	Ultra-fast, low latency, top-tier security
Sharma_Apartment_Guest	2.4 GHz	−68 dBm (Good / 3 bars)	WPA2-PSK (AES)	Slower speeds, standard security
Campus_Free_Public_WiFi	2.4 GHz	−84 dBm (Weak / 1-2 bars)	**Open (None / Unsecured)**	High packet loss, vulnerable to snooping

Network Selection & Justification:

- **Chosen Network: Home_Fiber_5G**
- **Technical Justification:**
    1. **Optimal Signal Strength:** At **−42 dBm-42\text{ dBm}−42 dBm**, the signal is strong (anything better than −50 dBm-50\text{ dBm}−50 dBm provides full link rate capacity and zero packet drop).
    2. **Superior Security:** It utilizes **WPA3-Personal** with Simultaneous Authentication of Equals (SAE), which protects against offline dictionary/brute-force attacks and ensures forward secrecy.
    3. **High Bandwidth:** Operating on the 5 GHz5\text{ GHz}5 GHz band provides wider channel widths with minimal interference compared to crowded 2.4 GHz2.4\text{ GHz}2.4 GHz frequencies.

Task 4: SSID Hiding in Wi-Fi Networks

What is 'SSID Hiding'?

- The **SSID (Service Set Identifier)** is the human-readable name of a Wi-Fi network.
- By default, a wireless access point continually broadcasts periodic **beacon frames** (101010 times per second) announcing its name so nearby devices can see it in their Wi-Fi menu.
- **SSID Hiding** disables the transmission of the network name in these beacon frames. The network is marked as "Hidden" or "Closed," meaning users must manually type the exact SSID name and password to connect.

One Major Advantage:

- **Protection Against Casual Users and Opportunistic Scanning:**

Because the network name does not appear in the standard device list, casual neighbors, passersby, and low-skilled attackers are less likely to attempt connecting or guessing passwords.

One Major Disadvantage:

- **False Sense of Security ("Security through Obscurity"):**

SSID hiding does not truly secure or hide the network from attackers. Free packet-sniffing tools (like *Wireshark* or *Aircrack-ng*) can detect the hidden SSID in seconds as soon as an authorized client device connects or reconnects, because the SSID is transmitted in plaintext inside **Probe Request** and **Probe Response** frames. Additionally, connecting manually creates extra setup hassle for legitimate users and guest devices.